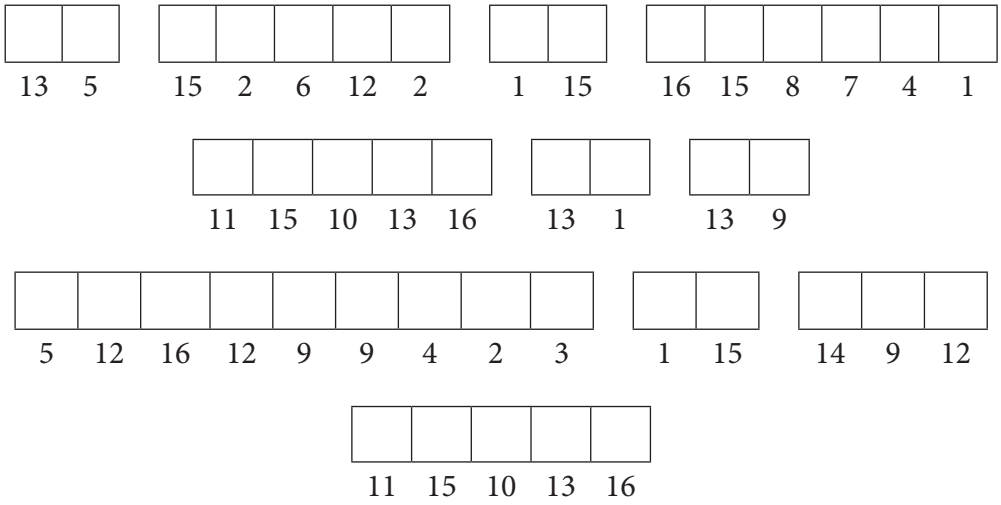
Name: _____

WORKSHEET

Factorising algebraic expressions

Logic is invincible because …



Pierre Boutroux

Rules

$ab + ac = a(b + c)$
$a^2 - b^2 = (a + b)(a - b)$
$ac + ad + bc + bd = (a + b)(c + d)$

© Cengage Learning Australia Pty Ltd 2013

Questions

Factorise the following algebraic expressions.

1 $10x + 25$

2 $3x^2 - 12x$

3 $-6y^2 - 2xy - 10y$

4 $x^2 - y^2$

5 $x^2 + 4x + 3$

6 $x^2 - 3x - 10$

7 $y^2 - 25$

8 $6x^2 + 5x - 4$

9 $y^2 + 4y + 4$

10 $4x^2 - 12x + 9$

11 $9x^2 - 1$

12 $12x^2 - 11x - 15$

13 $20x^2 + 64x - 21$

14 $25 - 49x^2$

15 $2x^2 + 7xy + 3y^2$

16 $12x^2 + 5xy - 2y^2$

Solutions

A $(x + y)(x - y)$

B $(y + 5)(y - 5)$

C $(3x + 2y)(4x - y)$

D $(x - 5)(x + 2)$

E $(4x + 3)(3x - 5)$

G $(2x - 3)^2$

I $(10x - 3)(2x + 7)$

L $(3x + 1)(3x - 1)$

M $(2x - 1)(3x + 4)$

N $(x + 3)(x + 1)$

O $(x + 3y)(2x + y)$

R $3x(x - 4)$

S $(y + 2)^2$

T $5(2x + 5)$

U $(5 + 7x)(5 - 7x)$

Y $-2y(3y + x + 5)$

© Cengage Learning Australia Pty Ltd 2013